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Course Code: CSE307

OBJECT ORIENTED ANALYSIS & DESIGN

Course Objectives

This course will help the learner to Infer, Analyse and Apply various techniques in Object Oriented Modeling, Analysis, Design and Implementation.

UNIT - I

15 Periods

Introduction: Definitions - OO Themes - Modeling - Abstraction - The three Models. **Class Modeling:** Object and Class concepts - Links and Associations - Generalization and Inheritance - a simple class model - Navigation of Class Models. **Advanced Class Modeling:** Concepts - Association ends - N-ary Association - Aggregation - Abstract class - Multiple Inheritance - Metadata - Reification - Constraints - Derived Data - Packages

UNIT - II

15 Periods

State Modeling: Events - States - Transitions and conditions - State diagrams - Behavior **Advanced State Modeling:** Nested State Diagrams - Signal Generalization - concurrency - Sample State Model - Relation of class and State Models. **Interaction Modeling:** Use Case Models - Sequence Models - Activity Models - Use case relationships - Special constructs for Activity Model

UNIT - III

15 Periods

Process Overview: Development Stages - Development Life Cycle - Devising a system Concept - Elaborating the concept - Preparing the Problem Statement. **Domain Analysis:** Overview - Domain Class Model - Domain State Model - Domain Interaction Model - Iterating the Analysis. **Application Analysis:** Application Interaction Model - Application Class Model - Application State Model - Adding Operations

UNIT - IV

15 Periods

System Design: Overview - Estimating Performance - Making a reuse Plan - Breaking the system into subsystems - Identifying concurrency - allocation of subsystems - Management of data storage - Handling global resources - common Architecture styles - Architecture of the ATM system. **Class Design:** Overview - Bridging the Gap - realizing Use Cases - Designing algorithms - Refactoring - Design Optimization - Organizing a Class Design - ATM Example. **Implementation Modeling:** Overview - fine-tuning classes - Fine-tuning Generalizations - Realizing Associations - Testing

TEXT BOOK

1. Michael R Blaha, James R Rumbaugh. *Object-Oriented Modeling and Design with UML*, Pearson, Second Edition, 2013.

REFERENCES

1. Ali Bahrami. *Object Oriented System Development*, McGraw-Hill Higher Education, 2015.
2. Grady Booch, Robert A. Maksimchuk Michael, W. Engle, Bobbi J. Young, Jim Conallen, and Kelli A. Houston. *Object-Oriented Analysis and Design with Applications*, Addison - Wesley, 2007.

ONLINE MATERIAL

1. <http://ocw.mit.edu/courses/architecture/4-297-special-problems-in-architecture-studies-fall-2000/labs/UML1.pdf>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	● Infer the basic concepts of Object-Orientation ● Infer and Create Class modelling
Unit II	● Infer the concepts of events, state, and transitions ● Create State diagrams, various kinds of Interaction diagrams
Unit III	● Infer the overview of Object-Oriented Development ● Perform Domain Analysis using various models ● Perform Application Analysis using various models
Unit IV	● Perform System Design and Class Design ● Infer various Implementation Techniques

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Infer the basic concepts and can create Class modelling
- Infer the concepts of events, state, transitions and interactions
- Create State diagrams and Interaction diagrams
- Perform Domain and Application Analysis
- Perform System Design and Class Design
- Infer various Implementation Techniques